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TOTAL
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number

NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2025

MATHEMATICS: PAPER II

EXAMINATION NUMBER

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Time: 3 hours

150 marks

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

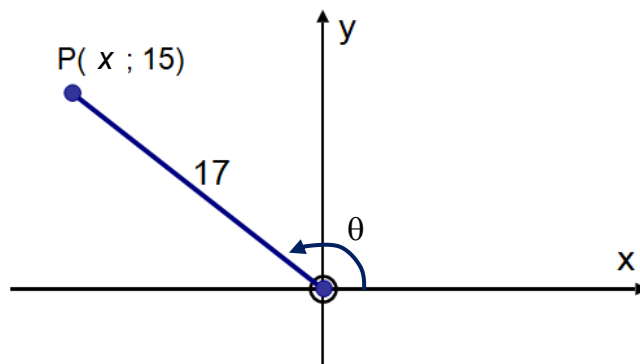
1. This question paper consists of 28 pages and an Information Sheet of 2 pages (i–ii). Please check that your question paper is complete.
2. Read the questions carefully.
3. **Answer ALL the questions on the question paper and hand it in at the end of the examination. Remember to write your examination number in the space provided on the question paper.**
4. Diagrams are not necessarily drawn to scale.
5. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
6. Ensure that your calculator is in **DEGREE** mode.
7. Clearly show **ALL** calculations, diagrams, graphs, etc. that you have used in determining your answers. **Answers only will NOT necessarily be awarded full marks.**
8. Round off to **TWO DECIMAL PLACES** unless otherwise stated.
9. It is in your own interest to write legibly and to present your work neatly.
10. One blank page (page 26) is included at the end of the question paper. Use this page if you run out of space for a question. Clearly indicate the number of the question you are answering, should you use this additional space. Spare graph papers are included on page 27 and 28.

FOR OFFICE USE ONLY: MARKER TO ENTER MARKS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14	TOTAL
7	9	11	7	6	14	13	7	11	10	16	12	16	11	150

SECTION A**QUESTION 1**

Given $17\sin\theta - 15 = 0$ and $\theta \in [90^\circ; 270^\circ]$.



(a) **Without** a calculator, determine the value of:

(1) $\tan(180^\circ + \theta)$

(3)

(2) $\cos(90^\circ + \theta)$

(2)

(b) **Use** a calculator to solve for θ .

(2)
[7]

QUESTION 2

(a) Showing all working, without the use of a calculator, determine the value of:

(1) $\tan 300^\circ$

(2)

(2) $\cos^2 15^\circ - \sin^2 15^\circ$

(2)

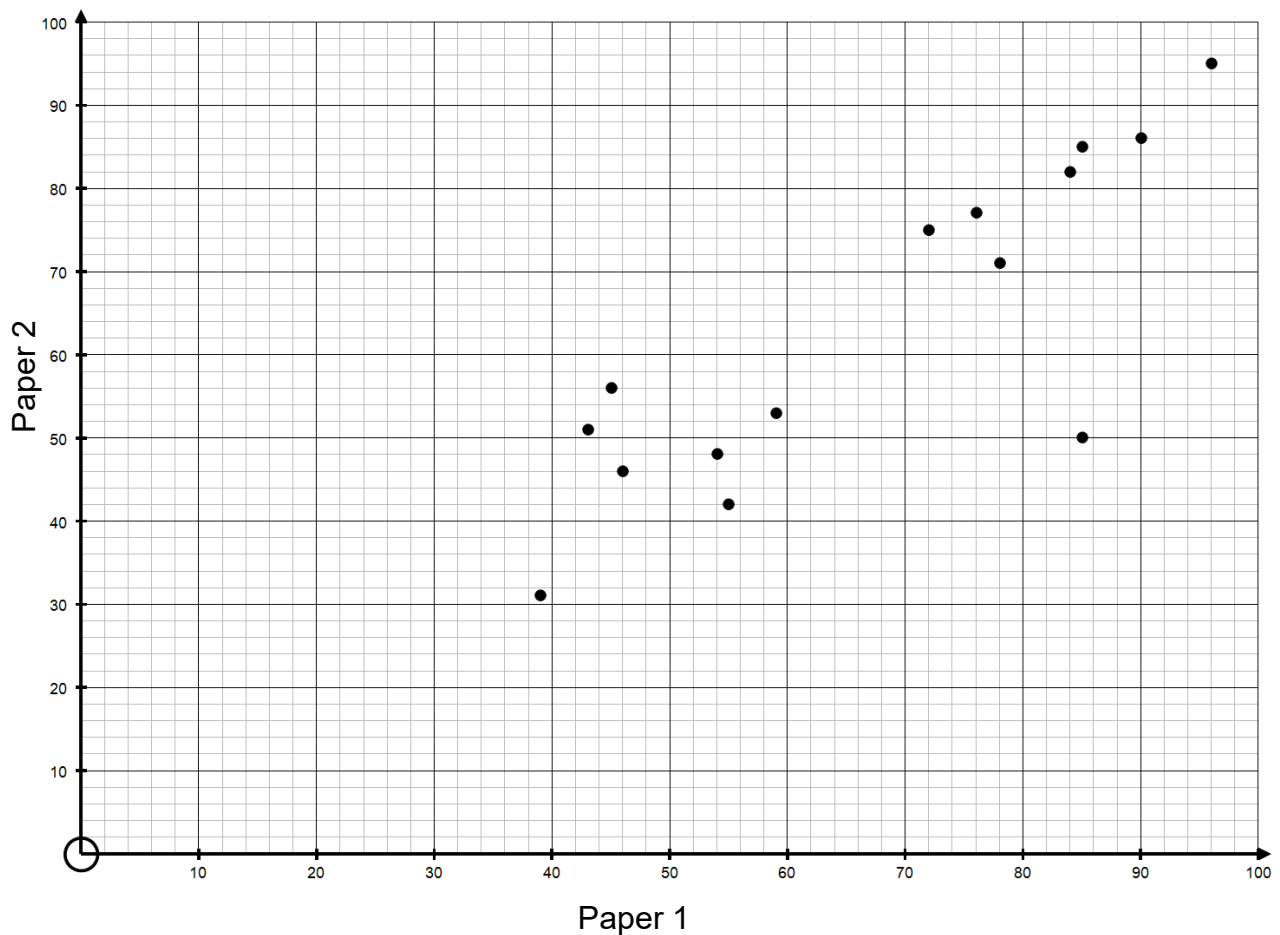
(b) Determine the general solution(s) for:

$$4\cos(2\theta + 60^\circ) = -2$$

(5)
[9]

QUESTION 3

- (a) The marks, as whole percentages, of 15 learners for their Grade 11 November Mathematics examinations are represented in the scatter plot below.



- (1) State the mark for Paper 1 of the learner who did the best.

(1)

- (2) State the lowest mark for Paper 2.

(1)

- (3) The line of best fit has the equation $y = 5,862 + 0,8541x$.

- (i) If a learner got 67% for Paper 1, what could that learner expect to get for Paper 2? Round your answer to the nearest whole.

(1)

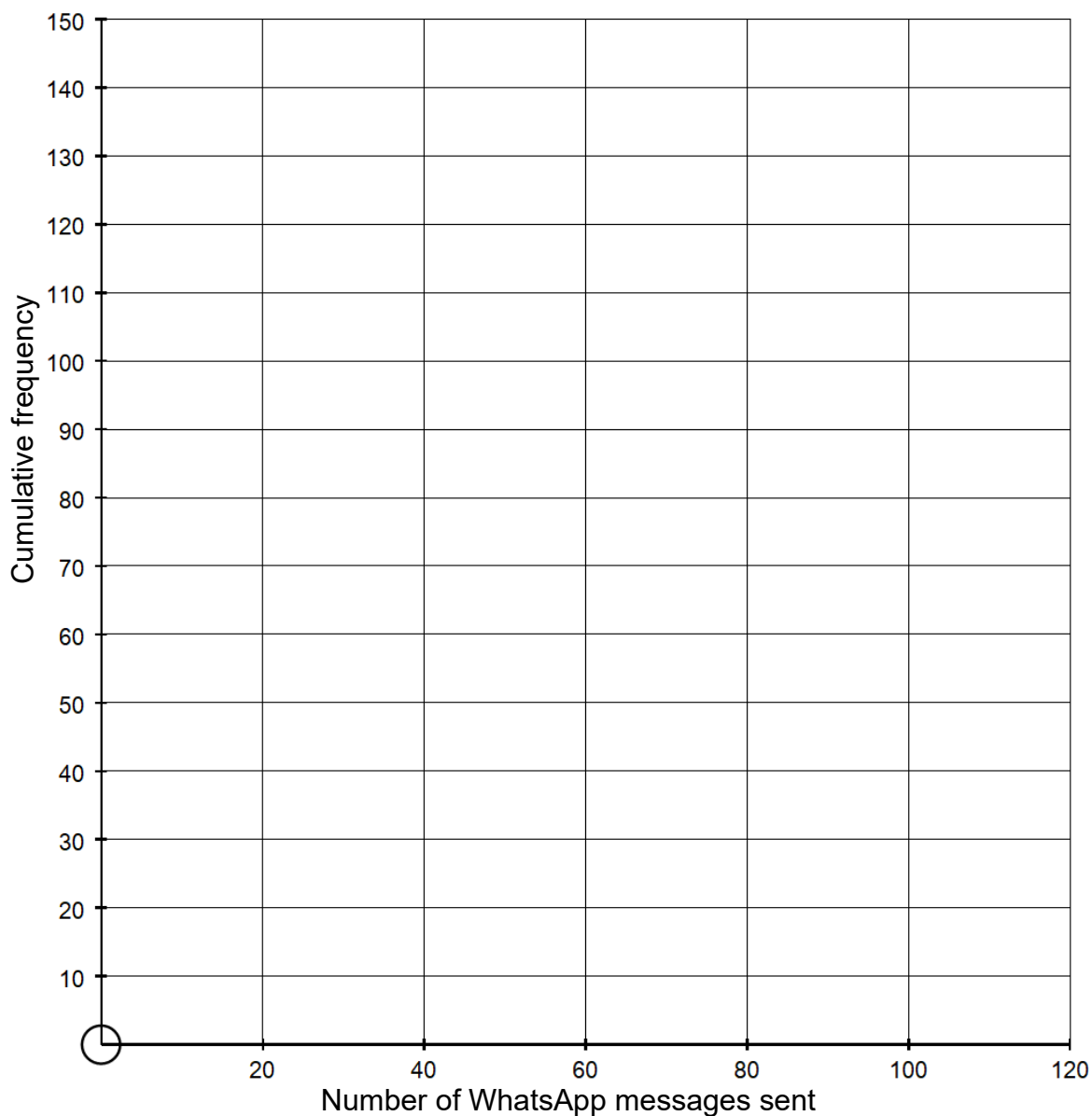
- (ii) If it is further given that the mean for Paper 1 ($\bar{x}$) was 67%. **On the scatter plot above**, sketch the line of best fit, clearly indicating the y-intercept and the point ($\bar{x}$; $\bar{y}$).

(3)

- (b) A survey of 140 people was done to see how many WhatsApp messages they sent in a day.

Complete the table of results and plot the ogive on the grid below.

Number of WhatsApp messages sent	Frequency	Cumulative frequency
$20 < x \leq 40$		14
$40 < x \leq 60$	32	
$60 < x \leq 80$	54	
$80 < x \leq 100$	27	
$100 < x \leq 120$		140



(5)
[11]

QUESTION 4

Sketched are the graphs $f(x) = a \cos x$ and $g(x) = \sin bx$ where $x \in [0^\circ; 210^\circ]$.



(a) State the amplitude of f .

(1)

(b) State the period of g .

(1)

(c) Write down the coordinates of:

(1) A, the turning point of g .

(1)

(2) B, the point of intersection of f and g .

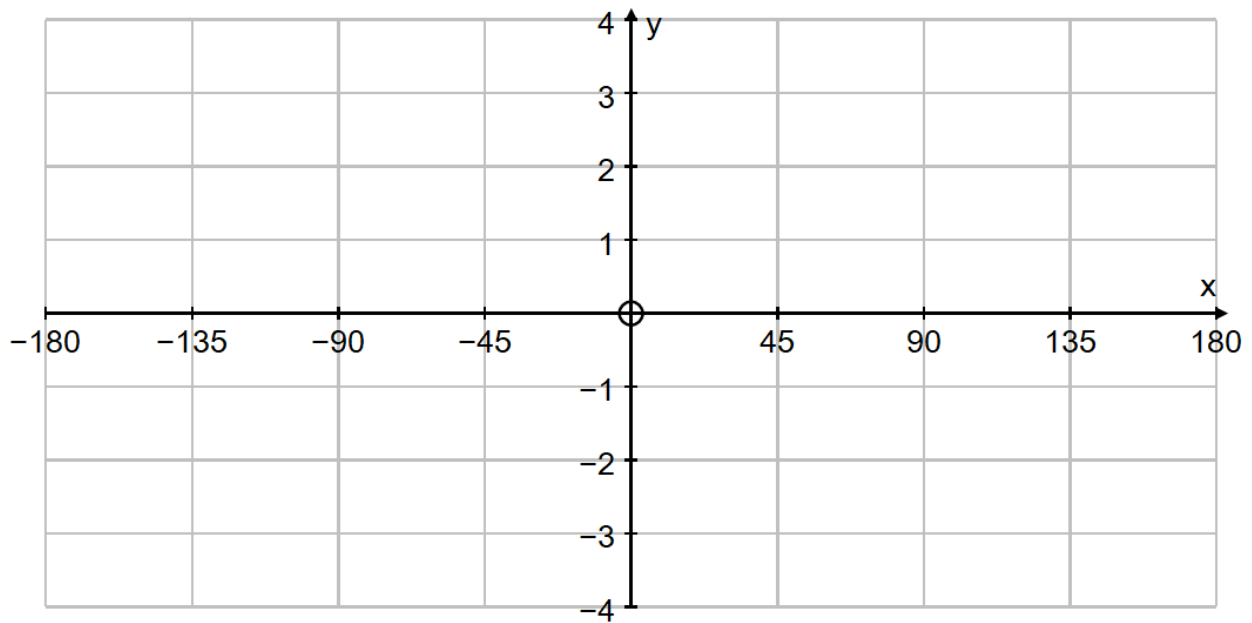
(1)

(d) If $x \in [0^\circ; 210^\circ]$, determine the value(s) of x where $g(x) \leq 0$.

(3)
[7]

QUESTION 5

- (a) On the axes below, sketch the graph of $h(x) = -\tan x$ where $x \in [-180^\circ; 180^\circ]$.



(3)

- (b) If $h(x)$ is shifted up two units. Write the equation of the new graph in the form $y = \dots$

(1)

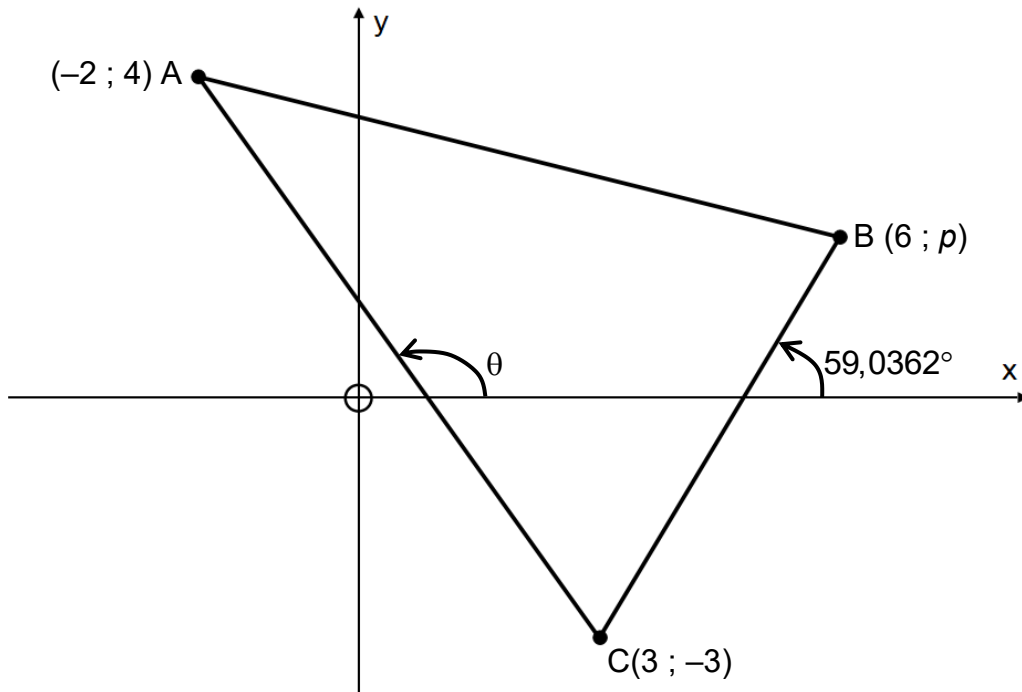
- (c) If $k(x) = -\tan(x - p)$. Give the value of p if the asymptotes of $k(x)$ are $x = -135^\circ$ and $x = 45^\circ$.

(2)
[6]

QUESTION 6

In the diagram:

- $\triangle ABC$ has vertices $A(-2 ; 4)$, $B(6 ; p)$ and $C(3 ; -3)$.
- BC has an angle of inclination of $59,0362^\circ$.



- (a) Find the gradient of AC .

(1)

- (b) Find θ and hence determine the size of angle $\hat{C}$.

(3)

- (c) Determine M, the midpoint of line AC.

(2)

- (d) Determine the equation of the line perpendicular to AC passing through the midpoint of AC (perpendicular bisector of AC). Leave your answer in the form $y = mx + c$.

(3)

- (e) Showing all working, determine the value of p .

(3)

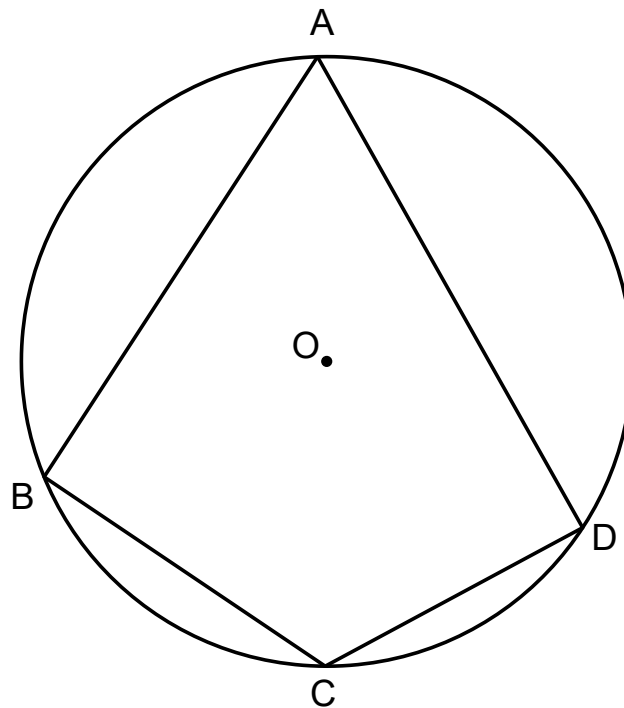
- (f) If the value of p is 2, determine D, the fourth point of the parallelogram ABCD. D lies in the third quadrant.

(2)
[14]

QUESTION 7

- (a) Use the diagram below to prove the statement that says:

'The opposite angles of a cyclic quadrilateral are supplementary.'



Required to prove: $\hat{A} + \hat{C} = 180^\circ$

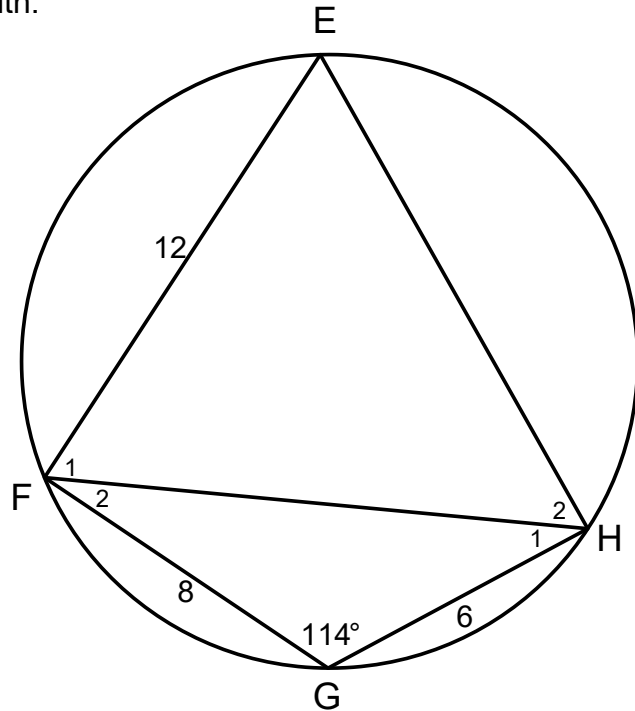
Construction(s): _____ (1)

Proof:

(4)

(b) Given cyclic quadrilateral EFGH with:

- $\hat{G} = 114^\circ$.
- $EF = 12$ units.
- $FG = 8$ units.
- $GH = 6$ units.



State all relevant reasons with your statements.

Determine:

(1) The length of FH.

(3)

(2) The size of $\hat{H}_2$, if $\hat{H}_2$ is an acute angle.

(5)
[13]

(b) Determine the length of GH.

(4)

(c) Give a reason why LM is the diameter of the circle passing through LHM.

(1)
[7]

74 marks

SECTION B**QUESTION 9**

(a) Prove:

$$\frac{\cos 65x \cos 63x + \sin 65x \sin 63x}{1 + \sin 2x} = \frac{\cos x - \sin x}{\cos x + \sin x}$$

(6)

- (b) Determine the values of x where $\frac{\cos 65x \cos 63x + \sin 65x \sin 63x}{1 + \sin 2x}$ is undefined, if $x \in [-90^\circ; 180^\circ]$.

QUESTION 10

- (a) At the beginning of a Grade 12 year, a survey was done to see how many community service hours each learner had already completed. The table below represents the findings.

Hours completed	Frequency (f)	Midpoint of class interval (x)	$f \cdot x$ (Frequency multiplied by midpoint)
$0 \leq h < 6$	5		
$6 \leq h < 12$	p		
$12 \leq h < 18$	28		
$18 \leq h \leq 24$	4		

It was estimated that the average number of hours already completed by the grade was 12.

By using the table, determine how many learners completed less than this average.

(b) A set of data has the properties listed below, where σ is the standard deviation and μ is the mean.

- $\sigma = 2$
- $\sum_{i=1}^n (x_i - \mu)^2 = 32$

(1) How many data values were used in this set of data?

(3)

(2) 10 new identical values are added to the data set; as a result, the mean remains the same. Determine the new standard deviation.

(2)
[10]

- (c) If the equation of line AE is $y = -\frac{4}{5}x + \frac{47}{5}$, determine the equation of the circle.

(7)

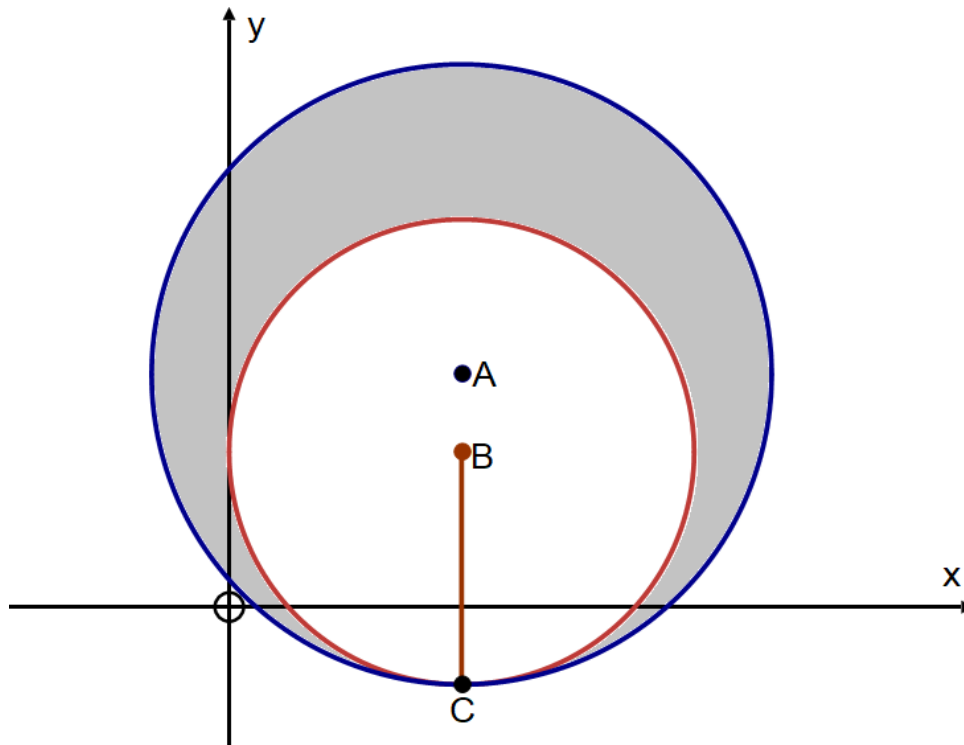
- (d) Determine the coordinates of B.

(4)
[16]

QUESTION 12

Given:

- Circle with centre A and circle with centre B.
- The two circles are tangent at point C.
- Circle with centre B is tangent to the y-axis.
- $BC \parallel y\text{-axis}$.
- A, B and C are co-linear.
- The equation of the circle with centre B is $x^2 + y^2 - 4(3x + 2y) = -16$.



- (a) Determine the equation of the tangent to the circle at C.

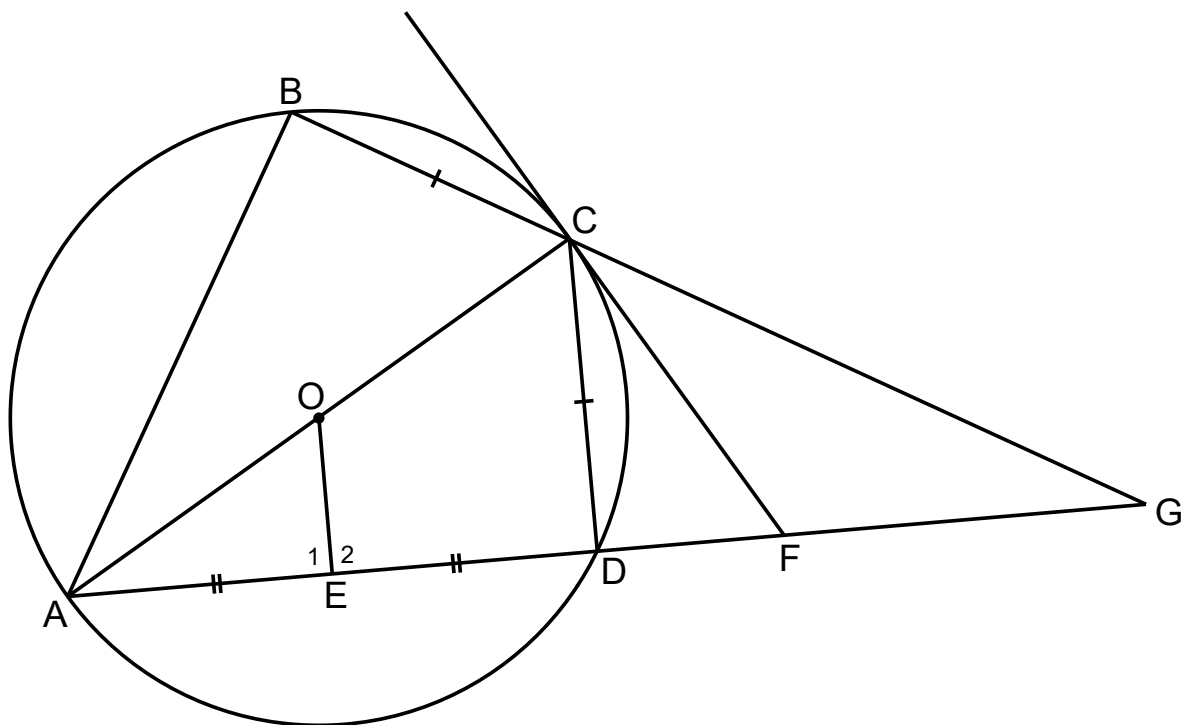
- (b) If the shaded region has an area of 28π units², determine the equation of the circle with centre A.

(6)
[12]

QUESTION 13

Given:

- O is the centre of the circle.
- CF is a tangent to the circle at C.
- $BC = CD$ and $AE = ED$.
- AD and BC are produced to G.



(a) Study the list of statements and reasons to answer the question below the table.

Statement	Reason
$\hat{C}_1 = \hat{C}_2 = x$	vertically opposite angles
$\hat{C}_1 = \hat{A}_2 = x$	tan chord theorem
$\hat{A}_2 + \hat{B} + \hat{C}_4 = 180^\circ$	interior angles of Δ
$\hat{A}_2 = \hat{A}_1 = x$	equal chords; equal angles in the same segment
$\hat{A}_2 + \hat{A}_1 + \hat{C}_4 + \hat{C}_5 = 180^\circ$	opposite angles of a cyclic quad
$\hat{C}_3 + \hat{C}_5 = 90^\circ$	tangent $\perp$ radius

Enter the correct **angle numbers** on the diagram **above** so that the statements and reasons make sense. (7)

State all relevant reasons with your statements.

(b) Prove that OEFC is a cyclic quadrilateral.

(4)

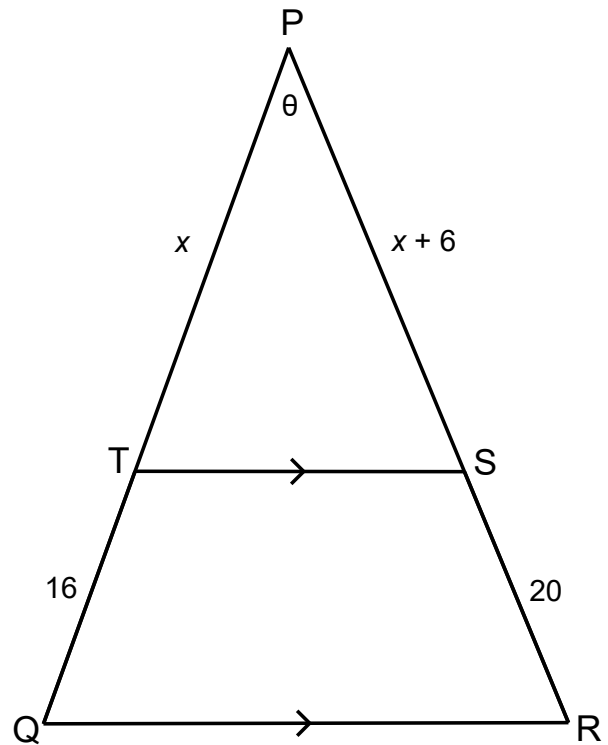
(c) If $\hat{GCF} = x$, prove that $CG^2 = AG \times FG$.

(5)
[16]

QUESTION 14

Given:

- $\triangle PQR$ with $TS \parallel QR$.
- $PT = x$ cm, $TQ = 16$ cm, $PS = x + 6$ cm, $SR = 20$ cm and $\hat{P} = \theta$.
- Area of $\triangle PTS = 258,96$ cm².

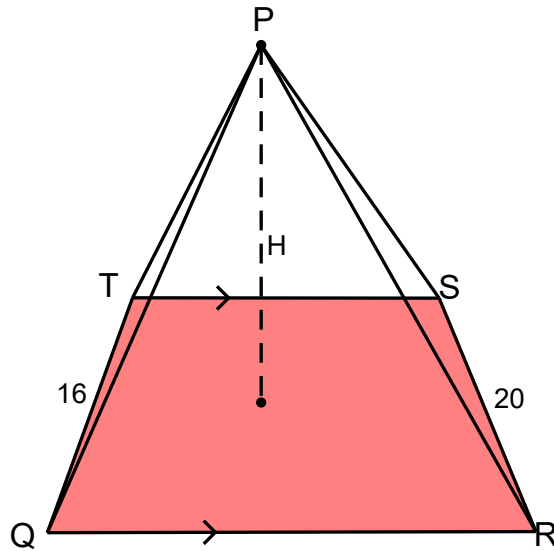


State all relevant reasons with your statements.

- (a) Determine the area of the trapezium $TQRS$.

(b) It is further given that:

- P is moved so that P is the apex of a trapezoidal pyramid with base TQRS.
- $\triangle PTS$ forms an angle of 60° with the base TQRS.



Determine the volume of the trapezoidal pyramid formed by making use of the formula $\text{Vol}_{\text{pyramid}} = \frac{1}{3} \text{area of base} \times H$.

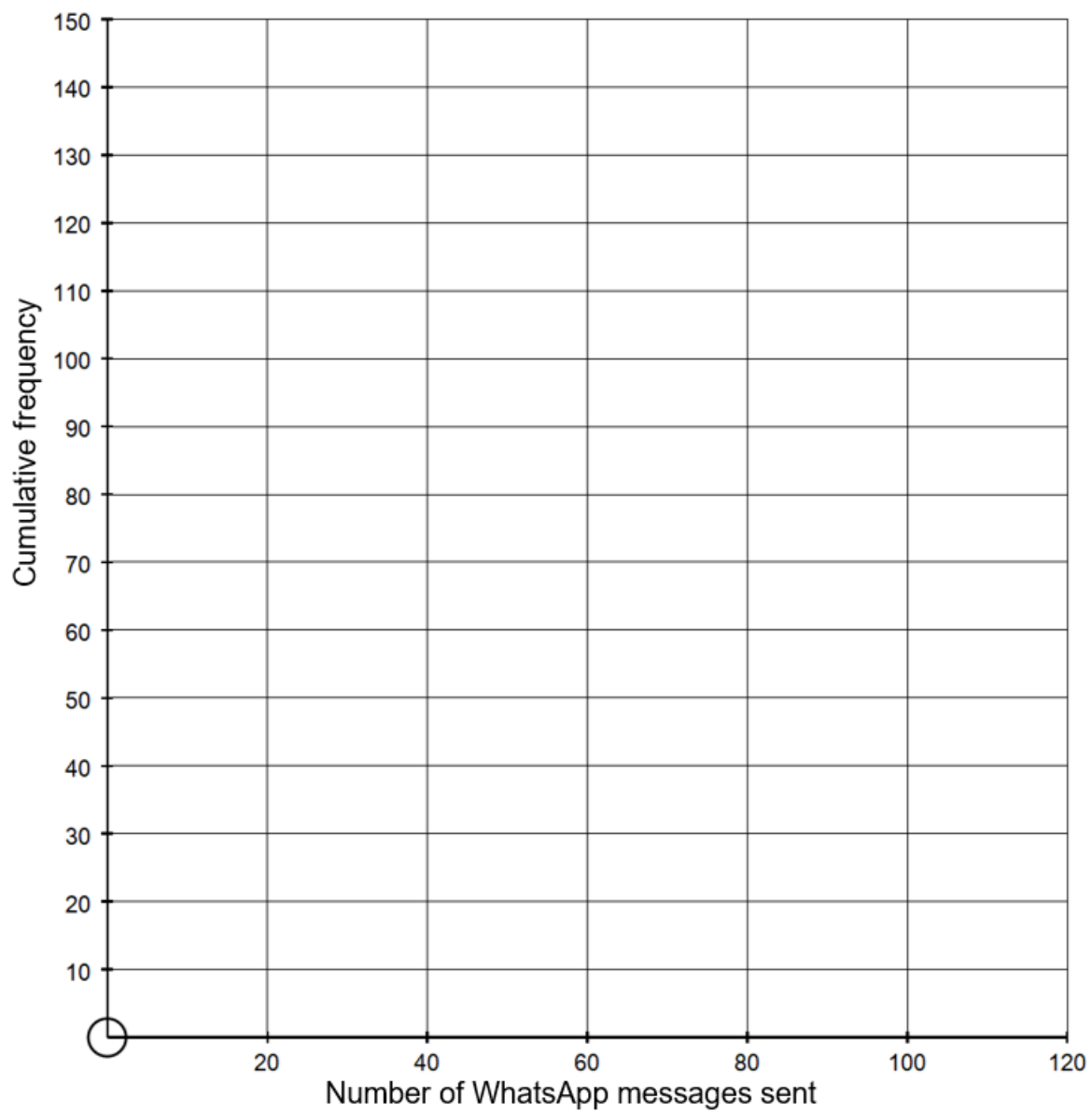
(5)
[11]

76 marks

Total: 150 marks

ADDITIONAL SPACE (ALL QUESTIONS)

**REMEMBER TO CLEARLY INDICATE AT THE QUESTION THAT YOU USED THE
ADDITIONAL SPACE TO ENSURE THAT ALL ANSWERS ARE MARKED.**

EXTRA GRAPHS**QUESTION 3 (b)**

QUESTION 5